SUSHIL RIJAL

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RESEARCH INTERESTS

Development economics; Public policy; Political economy; Infrastructure and institutions; Applied econometrics.

EDUCATION

The University of Texas at Arlington | Master of Science in Economic Data Analytics | GPA: 4.0 **Dec 2026**

- **Relevant Graduate Coursework:** Econometrics I and II; Time-Series Forecasting; Economics of Strategy; Monetary Policy and Financial System Analysis.
- **Fall 2026 Coursework (enrolled)**: Causal Inference; Advanced Economic Data Analytics; Mathematical Analysis.
- **Transferred coursework, University of Utah**: Manage Data for & with ML; Human-Centered Data Management covering SQL, NoSQL, Python, data wrangling, visualization, and data ethics.

Tribhuvan University, Nepal

M.A. in Economics | Patan Multiple Campus, Faculty of Humanities and Social Sciences **2019**

M.S. in Civil Engineering | Pulchowk Campus, Institute of Engineering **2019**

B.E. in Civil Engineering | Pulchowk Campus, Institute of Engineering **2015**

- **M.S. Thesis:** *Infrastructure Investment and Economic Growth in Nepal*, using Penn World Table data and an augmented Cobb-Douglas production framework.
- Awarded full-tuition entrance-merit scholarships as a top-2% entrant to the B.E. and M.S. Civil Engineering programs, along with additional merit scholarships and stipends.

RESEARCH EXPERIENCE

Anchor-Currency Volatility and Macroeconomic Transmission under a Peg | Independent Study May 2026 – July 2026

- Supervised by Prof. Aaron Smallwood; examines how Nepal's fixed parity with Indian rupee transmits third-currency volatility through prices, private credit, reserves, remittances, and trade composition.
- Built a reproducible workflow using Bayesian stochastic volatility, asymmetric GARCH models, realized variance and semivariance, local projections, parameter-stability tests, and uncertainty propagation.

Research Assistant | University of Utah Jan 2025 – Aug 2025

- Developed a reliability-based framework for combining noisy crowdsourced bounding-box annotations into higher-quality training targets for construction object-detection models.
- Supported the statistical and experimental evaluation of label integration across six construction and infrastructure datasets; the work was published in *Automation in Construction.*
- Conducted a statistical evaluation of the ROAD Alert real-time warning system and synthesized roadway-safety findings for a technical report.

METHODS & TOOLS

Econometrics: Panel data, difference-in-differences, instrumental variables, limited-dependent-variable and count-data models, nonlinear and interaction models, GMM, robustness checks, and model diagnostics.

Time Series: ARIMA and SARIMA, ARIMAX with Fourier terms, NNETAR, VAR and VECM, unit-root and cointegration analysis, forecasting, impulse-response analysis, structural-stability testing, and volatility modeling.

Economic & Policy Analysis: Cost-benefit analysis**,** discounted cash flow, NPV and IRR, scenario and sensitivity analysis.

Programming & Data: Python, Stata, R, MATLAB, EViews, SQL, Excel, Tableau, and Power BI.

Machine Learning: Random Forest, XGBoost, PCA, SHAP, data wrangling, feature engineering, and cross-validation.

PUBLICATIONS

- Golazad, S., **Rijal, S**., Asgari, S., Medina, J. C., & Rashidi, A. (2026). "Feature-informed reliability-based merging of crowdsourced labels for digital construction object detection." *Automation in Construction*, 188, 107031.

ADDITIONAL QUANTITATIVE PREPARATION

- **Deep Learning,** University of Utah, Grade A; **Microeconomics, MITx,** Grade A (96%)**; GRE Quantitative**, 170/170
- Selected Certificates: Macroeconomic Diagnostics, IMFx; Network Analysis and Economic Complexity for Sustainable Development, UNU-MERIT; Unlocking Investment and Finance in Emerging Markets and Developing Economies, WBGx.

SELECTED POLICY, INFRASTRUCTURE & CONSULTING EXPERIENCE

Economic Feasibility Consultant, Project-Based | Bardibas Medical College *2021*

- Developed a discounted cash-flow and socioeconomic feasibility model incorporating NPV, IRR, scenario and sensitivity analysis, social benefit-cost ratios, and the economic rate of return to inform the investment decision.

Team Leader | United Nations Office for Project Services *Jan 2019 – Apr 2019*

- Supervised seismic-retrofitting work on model public buildings, coordinated field teams and contractors, and monitored cost-effectiveness and implementation quality.

Reconstruction Engineer | Government of Nepal, Rasuwa *Apr 2016 – Dec 2016*

- Served as a government field representative for post-earthquake reconstruction, coordinating households, contractors, local officials, and implementation activities in a remote mountain district.

Lead Trainer and Consultant, Project-Based | Nepal *Jan 2017 – Oct 2023*

- Designed and delivered disaster-resilient construction training to more than 1,000 engineers, sub-engineers, and masons across earthquake-affected districts and municipalities in Bagmati, Gandaki, and Koshi Provinces.

SELECTED PROJECTS

Life Expectancy, Development, and the Preston Curve

- Analyzed World Development Indicators for 185 countries from 2000 to 2023 to examine how income, health systems, and infrastructure relate to life expectancy across stages of development.
- Engineered composite indices and nonlinear and interaction specifications; validated results with two-way fixed effects and extended the analysis to a pre-COVID high-income-country sample; produced reproducible Stata code, a technical report, and policy-oriented visualizations.

Data Integration and GDP Prediction using Open Data Sources

- Integrated World Bank, IMF, UNDP, and related indicators in Python; built a documented pipeline for data cleaning, harmonization, feature engineering, and multicollinearity assessment.
- Compared fixed- and random-effects Cobb-Douglas specifications with Random Forest and XGBoost models; used PCA for dimensionality reduction and SHAP for model interpretation.

Texas Electricity Generation Forecasting

- Built forecasting models in R and selected SARIMA specifications using unit-root tests, BIC and AICc criteria, STL decomposition, and a Box-Cox transformation selected using Guerrero’s method.
- Validated final SARIMA model via Ljung-Box residual testing, achieving a 2.8% MAPE on a 12-month holdout set.

ACADEMIC EXPERIENCE

Graduate Teaching Assistant | The University of Texas at Arlington Sep 2025 – Present

- Evaluated microeconomics assignments and provided targeted feedback to strengthen students’ analytical skills.
- Enhanced accessibility and clarity of digital instructional materials to improve student learning outcomes.

Graduate Teaching Assistant | University of Utah Aug 2024 – Dec 2024

- Graded undergraduate civil and environmental engineering coursework and provided written feedback on technical problem solving.

Lecturer | Madan Bhandari College of Engineering and Kantipur Engineering College, Nepal Apr 2019 – May 2024

- Taught engineering economics, project evaluation, demand forecasting, hypothesis testing, and quantitative and qualitative research methods.
- Led a collaborative faculty research study comparing the life-cycle costs of diesel and electric bus fleets, including long-term operating costs, energy inputs, and investment trade-offs.

HONORS, AWARDS, AFFILIATIONS & LEADERSHIP

- Wayne Watts Graduate Fellowship (2026), McCrea Economics Scholarship (2025, 2026), UT Arlington.
- Beta Gamma Sigma, International Business Honor Society, inducted 2026.
- Best Paper Award, "Life-Cycle Costing Comparison of Diesel and Electric Bus," KEC Transportation Symposium (2019), 2nd International Conference.
- Founder and Digital Community Lead, MS/PhD Nepali Aspirants and GRE Study Nepal, peer-mentoring and resource-sharing communities serving 89,000+ members.